

Introduction to Mathematics and Optimization

May, 2022

The references in the suggested solutions to this examination paper are with respect to the PDF edition *Introduction to Mathematics and Optimization* (Compressed static version (15.11.2021) of interactive book).

PROBLEM 1

Let $f : \mathbb{R}^2 \rightarrow \mathbb{R}$ be the function given by

$$f(x, y) = x + y^2$$

and

$$C = \{(x, y) \in \mathbb{R}^2 \mid x \leq 1, y \leq 1, x + 2y \geq 1\}.$$

In this problem we focus on the optimization problem

$$\begin{aligned} \min f(x, y) \\ (x, y) \in C. \end{aligned} \tag{1}$$

(a) Show that (1) is a convex optimization problem.

ANSWER:

The Hessian matrix of f is $\begin{pmatrix} 0 & 0 \\ 0 & 2 \end{pmatrix}$. This matrix is positive semidefinite, since it is a diagonal matrix with non-negative entries (Exercise 8.26). Hence f is a convex function by Theorem 8.21. The subset C is an intersection of three convex sets of the form $\{v \in \mathbb{R}^d \mid f(v) \leq a\}$, where $f : \mathbb{R}^d \rightarrow \mathbb{R}$ is convex (Lemma 4.26). In the present case, each of these functions has a Hessian matrix with zero entries (again concluding by Theorem 8.21). Thus C is convex, since it is an intersection of three convex subsets (Exercise 4.22). Therefore (1) is a convex optimization problem.

(b) Explain without drawing why C is a compact set and why this implies that (1) has a solution.

ANSWER:

A set being compact means that it is closed and bounded according to Definition 5.75. C is closed, since it is an intersection of three closed sets of the form $\{v \in \mathbb{R}^d \mid f(v) \leq a\} = f^{-1}((-\infty, a])$, where $f : \mathbb{R}^d \rightarrow \mathbb{R}$ is a continuous function (Proposition 5.72). The continuity of these functions follows from Lemma 5.62 and Proposition 5.64. Furthermore, $(-\infty, a]$ is a closed subset of $\mathbb{R}$ by Proposition 5.51. That C is bounded follows from applying Fourier-Motzkin elimination: $x \leq 1$ and $x + 2y \geq 1$ implies $1 \geq 1 - 2y$ and hence $y \geq 0$. Similarly, $1 \geq y \geq \frac{1}{2} - \frac{1}{2}x$ implies $x \geq -1$. Thus $(x, y) \in C \implies x^2 + y^2 \leq 1^2 + 1^2 = 2$, and C is bounded according to §5.2.3. The optimization problem has a solution by Theorem 5.76, since C is compact and $f(x, y) = x + y^2$ is continuous, as it is convex and defined on an open set (Theorem 6.6).

(c) Solve the optimization problem (1) using a method from the curriculum.

ANSWER:

We use the method described in §7.10. The critical points of f are given by

$$\begin{aligned} \frac{\partial f}{\partial x} &= 1 = 0 \\ \frac{\partial f}{\partial y} &= 2y = 0 \end{aligned}$$

That is, there are no critical points for f . We examine optima on the boundary of C . This is given by

$$\partial C = \{(x, y) \in C \mid x = 1\} \cup \{(x, y) \in C \mid y = 1\} \cup \{(x, y) \in C \mid x + 2y = 1\}.$$

On the first part above, the optimization problem is $\min 1 + y^2$ with the constraint $0 \leq y \leq 1$. Here the minimum is 1 and is attained for $y = 0$. On the middle part above, the optimization problem is $\min 1 + x$ with the constraint $-1 \leq x \leq 1$. Here the minimum is 0 and is attained for $x = -1$. On the last part, the optimization problem is $\min g(y) = 1 - 2y + y^2$ with the constraint $0 \leq y \leq 1$. Here $g'(y) = 2y - 2$ and the minimum 0 is at $y = 1$, since $g'(1) = 0$ and the parabola $g(y)$ opens upward. Overall, the minimum for (1) is 0 and is attained at $(-1, 1)$.

- (d) Write down the KKT conditions corresponding to (1).

ANSWER:

According to (9.24), these are

$$\begin{aligned}\lambda_1, \lambda_2, \lambda_3 &\geq 0 \\ \lambda_1(x - 1) &= 0 \\ \lambda_2(y - 1) &= 0 \\ \lambda_3(1 - x - 2y) &= 0 \\ 1 + \lambda_1 - \lambda_3 &= 0 \\ 2y + \lambda_2 - 2\lambda_3 &= 0\end{aligned}$$

- (e) Explain solely from the KKT conditions why $(1, 1)$ cannot be an optimal solution to (1).

ANSWER:

First, we note that (1) is strictly feasible, since $(x, y) = (\frac{1}{2}, \frac{1}{2})$ satisfies the strict inequalities $x < 1, y < 1, x + 2y > 1$. Theorem 9.24(i) therefore states that if $(1, 1)$ is an optimal solution, then the KKT conditions can be solved for $x = 1$ and $y = 1$. We disprove this: Since $1 - x - 2y < 0$ for $(x, y) = (1, 1)$, we must have $\lambda_3 = 0$ and hence $1 + \lambda_1 = 0$, which contradicts $\lambda_1 \geq 0$.

- (f) Is f strictly convex?

ANSWER:

f is not strictly convex, since $t = f((1 - t)u + tv) = (1 - t)f(u) + tf(v) = t$ for $u = (0, 0), v = (1, 0)$ and $t \in \mathbb{R}$ according to Definition 6.1.

- (g) Give an example of a convex optimization problem of the form $\min_{x \in A} h(x)$ with $h : \mathbb{R} \rightarrow \mathbb{R}$ and $A \subseteq \mathbb{R}$, which does not have a solution, but where A is bounded.

ANSWER:

An example is $h(x) = x$ and $A = (0, 1]$. If $x_0 \in A$, then $A \ni \frac{1}{2}x_0 < x_0$. That is, an optimum does not exist.

PROBLEM 2

Let $f : \mathbb{R}^2 \rightarrow \mathbb{R}$ be the function given by

$$f(x, y) = x^2 + xy + y^3$$

and

$$C = \{(x, y) \in \mathbb{R}^2 \mid x \leq 2, y \leq 2, x + y \geq 3\}.$$

In this problem we focus on the optimization problem

$$\begin{aligned} \min f(x, y) \\ (x, y) \in C. \end{aligned} \tag{2}$$

(a) Find the critical points of f .

ANSWER:

The gradient of f is

$$\nabla(f) = \left(\frac{\partial f}{\partial x}, \frac{\partial f}{\partial y} \right) = (2x + y, x + 3y^2)$$

The critical points are the solutions (according to Definition 7.17) to the equations

$$\begin{aligned} 2x + y &= 0 \\ x + 3y^2 &= 0, \end{aligned}$$

With $y = -2x$, substitution into $x + 3y^2 = 0$ gives $x + 12x^2 = 0$. We therefore obtain the critical points $(0, 0)$ and $(-\frac{1}{12}, \frac{1}{6})$.

(b) Does f have a local minimum?

ANSWER:

The Hessian matrix of f is

$$\begin{pmatrix} 2 & 1 \\ 1 & 6y \end{pmatrix}$$

Substituting the point $(-\frac{1}{12}, \frac{1}{6})$ gives the matrix

$$\begin{pmatrix} 2 & 1 \\ 1 & 1 \end{pmatrix},$$

which is positive definite according to Exercise 3.42. By Theorem 8.9(i), this point is thus a local minimum.

(c) Show without a drawing that $x \geq 1$ and $y \geq 1$ if $(x, y) \in C$.

ANSWER:

For $(x, y) \in C$, we have $x \leq 2$ and $3 - y \leq x$. This gives $3 - y \leq 2$ and hence $y \geq 1$. Similarly, $x \geq 1$ is seen by comparing $y \leq 2$ and $3 - x \leq y$.

(d) Show that f is not a convex function.

ANSWER:

We have computed the Hessian matrix of f in part (b). At the point $(0, 0)$, the Hessian matrix is

$$\begin{pmatrix} 2 & 1 \\ 1 & 0 \end{pmatrix}$$

which is not positive semidefinite, since it transforms to the diagonal matrix

$$\begin{pmatrix} 2 & 0 \\ 0 & -\frac{1}{2} \end{pmatrix}$$

via the operations in §8.6. The conclusion follows from Exercise 8.26 and Proposition 3.41. Hence the function is not convex according to Theorem 8.21.

- (e) Show that f is a strictly convex function when considered as a function on C .

ANSWER:

The function is strictly convex when $y > \frac{1}{12}$ according to Exercise 3.42. Hence f is strictly convex on C according to Theorem 8.21.

- (f) Show that the optimization problem (2) has a unique solution.

ANSWER:

The optimization problem has a solution, since C is compact and f is continuous (Theorem 5.76). C is bounded, because $(x, y) \in C \implies x^2 + y^2 \leq 2^2 + 2^2 = 8$, since $1 \leq x \leq 2$ and $1 \leq y \leq 2$ if $(x, y) \in C$ according to §5.2.3. C is closed, since it is an intersection of three closed sets of the form $\{v \in \mathbb{R}^d \mid g(v) \leq a\} = g^{-1}((-\infty, a])$, where $g : \mathbb{R}^d \rightarrow \mathbb{R}$ is a continuous function (Proposition 5.72). The continuity of these functions follows from Lemma 5.62 and Proposition 5.64. Furthermore, $(-\infty, a]$ is a closed subset of $\mathbb{R}$ by Proposition 5.51. The same argument shows that f is continuous. Hence Theorem 5.76 can be applied. A solution to the optimization problem is unique because f is strictly convex (Theorem 6.12).

- (g) Solve the optimization problem (2).

ANSWER:

The optimum cannot lie in C° , since it would then have to be a critical point according to Proposition 7.44. There are no critical points of f in C . We therefore examine the boundary ∂C , which consists of the $(x, y) \in C$ with $x = 2$ or $y = 2$ or $x + y = 3$. For $x = 2$, we obtain the optimization problem $\min y^3 + 2y + 4$ for $y \in [1, 2]$. Since $3y^2 + 2 = 0$ has no solutions, there are no critical points. However, there is a minimum at $y = 1$ with value 7. For $y = 2$, we obtain the optimization problem $\min x^2 + 2x + 8$ for $x \in [1, 2]$. Here the minimum is at $x = 1$ with value 11. For $x + y = 3$, we substitute $x = 3 - y$ into $f(x, y) = x^2 + xy + y^3$ and obtain the optimization problem $\min y^3 - 3y + 9$ for $y \in [1, 2]$. There are critical points where $3y^2 - 3 = 0$, but these fall outside $[1, 2]$. The minimum is at $y = 1$ with value 7. That is, the minimum here is at $(2, 1)$.

Overall, the smallest value attained on the boundary is 7, and it is attained at $(2, 1)$, which is thus the solution to (2).

PROBLEM 3

- (a) Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be the function given by

$$f(x) = x^4 + x^3 + x^2 + x + 1.$$

Show that f is strictly convex.

ANSWER:

We compute

$$f''(x) = 12x^2 + 6x + 2.$$

The discriminant of the above quadratic polynomial is $6^2 - 4 \cdot 12 \cdot 2 = -60$. Hence $f''(x) > 0$ for all $x \in \mathbb{R}$, and Corollary 6.47 implies that f is strictly convex.

- (b) Let $g : \mathbb{R} \rightarrow \mathbb{R}$ be given by

$$g(x) = \frac{1}{12}x^4 - \frac{5}{6}x^3 + 3x^2 + x + 1.$$

Show that g is not convex.

ANSWER:

We compute

$$g''(x) = x^2 - 5x + 6.$$

The quadratic polynomial above has roots at 2 and 3. The corresponding parabola opens upward. Thus $g''(x) < 0$ for $x \in (2, 3)$. Hence g cannot be convex, again by Corollary 6.47.

- (c) The previous part illustrates that g as a function on all of $\mathbb{R}$ is not convex. Give two non-empty convex open subsets $U_1, U_2 \subseteq \mathbb{R}$ with $U_1 \cap U_2 = \emptyset$, such that $-10 \in U_1$, $10 \in U_2$, and g is convex when considered as a function on U_1 and U_2 .

ANSWER:

According to the solution above, we have $g''(x) \geq 0 \iff x \leq 2 \vee x \geq 3$. We can therefore choose $U_1 = (-\infty, 2)$ and $U_2 = (3, \infty)$. Since $\mathbb{R} \setminus U_1 = [2, \infty)$ is closed by Proposition 5.51, U_1 is open by Definition 5.53. Similarly for U_2 . Arbitrary intervals are seen to be convex analogously to Exercise 4.21. It follows again from Corollary 6.47 that $g : U_1 \rightarrow \mathbb{R}$ and $g : U_2 \rightarrow \mathbb{R}$ are convex (indeed strictly convex).